A result of Bui–Pratt–Zaharescu, and Erdös problem #437

Terence Tao · 9 August 2024

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The following problem was posed by Erdös and Graham (and is listed as problem #437 on the Erdös problems website):

> **Problem 1** Let $1 \leq a_1 < \ldots < a_k \leq x$ be integers. How many of the partial products a_1, $a_1 a_2$, ..., $a_1 \ldots a_k$ can be squares? Is it true that, for any $\varepsilon > 0$, there can be more than $x^{1-\varepsilon}$ squares?

If one lets $L(x)$ denote the maximal number of squares amongst such partial products, it was observed in the paper of Erdös and Graham that the bound $L(x) = o(x)$ is "trivial" (no proof was provided, but one can for instance argue using the fact that the number of integer solutions to hyperelliptic equations of the form $(n+h_1)\ldots(n+h_d) = m^2$ for fixed $h_1 < \ldots < h_d$ is quite sparse, and in fact finite for $d > 2$ thanks to Siegel's theorem), and the problem then asks if $L(x) = x^{1-o(1)}$.

It turns out that this problem was essentially solved (though not explicitly) by a recently published paper of Bui, Pratt, and Zaharescu, who studied a closely related quantity t_n introduced by Erdös, Graham, and Selfridge (see also Problem B30 of Guy's book), defined for any natural number n as the least natural number t_n such that some subset of $n+1, \ldots, n+t_n$, when multiplied together with n, produced a square. Among the several results proven about t_n in that paper was the following:

> **Theorem 2 (Bui–Pratt–Zaharescu, Theorem 1.2)** For x sufficiently large, there exist $\gg x \exp(-(3\sqrt{2}/2 + o(1))\sqrt{\log x}\sqrt{\log\log x})$ integers $1 \leq n \leq x$ such that $t_n \leq \exp((\sqrt{2} + o(1))\sqrt{\log x}\sqrt{\log\log x})$.

The arguments were in fact quite elementary, with the main tool being the theory of smooth numbers (the theory of hyperelliptic equations is used elsewhere in the paper, but not for this particular result).

If one uses this result as a "black box", then an easy greedy algorithm argument gives the lower bound

$$L(x) \geq x \exp(-(5\sqrt{2}/2 + o(1))\sqrt{\log x}\sqrt{\log\log x}),$$

but with a small amount of additional work, one can modify the *proof* of the theorem to give a slightly better bound:

> **Theorem 3 (Bounds for L)** As $x \to \infty$, we have the lower bound
>
> $$L(x) \geq x \exp(-(\sqrt{2} + o(1))\sqrt{\log x}\sqrt{\log\log x})$$
>
> and the upper bound
>
> $$L(x) \leq x \exp(-(1/\sqrt{2} + o(1))\sqrt{\log x}\sqrt{\log\log x}).$$
>
> In particular, for any $\varepsilon > 0$, one has $L(x) \geq x^{1-\varepsilon}$ for sufficiently large x.

The purpose of this blog post is to record this modification of the argument, which is short enough to present immediately. For a large x, let u denote the quantity

$$u := \sqrt{2}\sqrt{\log x}/\sqrt{\log\log x}.$$

We call a natural number $x^{1/u}$-*smooth* if all of its prime factors are at most $x^{1/u}$. From a result of Hildebrand (or the older results of de Bruijn), we know that the number $\pi(x, x^{1/u})$ of $x^{1/u}$-smooth numbers less than or equal to x is

$$\begin{aligned}\pi(x, x^{1/u}) &= x \exp(-(1+o(1))u \log u) \\ &= x \exp(-(1/\sqrt{2}+o(1))\sqrt{\log x}\sqrt{\log\log x}).\end{aligned} \tag{1}$$

Let $\pi(x^{1/u})$ be the number of primes up to $x^{1/u}$. From the prime number theorem we have

$$\begin{aligned}\pi(x^{1/u}) &= (1+o(1))x^{1/u}/\log x^{1/u} \\ &= \exp((1/\sqrt{2}+o(1))\sqrt{\log x}\sqrt{\log\log x}).\end{aligned} \tag{2}$$

To prove the lower bound on $L(x)$, which is a variant of Theorem 2. The key observation is that given any $\pi(x^{1/u})+1$ $x^{1/u}$-smooth numbers $b_1, ..., b_{\pi(x^{1/u})+1}$, some non-trivial subcollection of them will multiply to a square. This is essentially Lemma 4.2 of Bui–Pratt–Zaharescu, but for the convenience of the reader we give a full proof here. Consider the multiplicative homomorphism $f : \mathbf{N} \to (\mathbf{Z}/2\mathbf{Z})^{\pi(x^{1/u})}$ defined by

$$f(n) := (\nu_{p_i}(n) \bmod 2)_{i=1}^{\pi(x^{1/u})},$$

where p_i is the ith prime and $\nu_{p_i}(n)$ is the number of times p_i divides n. The vectors $f(b_1), ..., f(b_{\pi(x^{1/u})+1})$ lie in a $\pi(x^{1/u})$-dimensional vector space over $\mathbf{Z}/2\mathbf{Z}$, and thus are linearly dependent. Thus there exists a non-trivial collection of these vectors that sums to zero, which implies that the corresponding elements of the sequence $b_1, ..., b_{\pi(x^{1/u})+1}$ multiply to a square.

From (1), (2) we can find $x \exp(-(\sqrt{2}+o(1))\sqrt{\log x}\sqrt{\log\log x})$ sequences of $x^{1/u}$-smooth numbers $b_1 < ... < b_{\pi(x^{1/u})+1}$ in $\{1, ..., x\}$, with each sequence being to the right of the previous sequence. By the above observation, each sequence contains some non-trivial subcollection that multiplies to a square. Concatenating all these subsequences together, we obtain a single sequence $1 \leq a_1 < ... < a_k \leq x$ with at least $x \exp(-(\sqrt{2}+o(1))\sqrt{\log x}\sqrt{\log\log x})$ partial products multiplying to a square, giving the desired lower bound on $L(x)$.

Next, we prove the upper bound on $L(x)$. Suppose that a sequence $1 \leq a_1 < ... < a_k \leq x$ has $L(x)$ partial products $a_1...a_{i_l}$ that are squares for some $1 \leq i_1 < ... < i_{L(x)} \leq k$. Then we have $a_{i_l+1}...a_{i_{l+1}}$ a square for all $0 \leq l < L(x)$ (with the convention $i_0 = 0$). The key observation (essentially Lemma 3.4 of Bui–Pratt–Zaharescu) is that, for each $0 \leq l < L(x)$, one of the following must hold:

- (i) At least one of the $a_{i_l+1}, ..., a_{i_{l+1}}$ is $x^{1/u}$-smooth.
- (ii) At least one of the $a_{i_l+1}, ..., a_{i_{l+1}}$ is divisible by p^2 for some prime $p > x^{1/u}$.
- (iii) $a_{i_{l+1}} - a_{i_l+1} > x^{1/u}$.

Indeed, suppose that (i) and (ii) are not true, then one of the terms in the sequence $a_{i_l+1}, ..., a_{i_{l+1}}$ is divisible by exactly one copy of p for some prime $p > x^{1/u}$. In order for the product $a_{i_l+1}...a_{i_{l+1}}$ to be a square, another element of the sequence must also be divisible by the same prime; but this implies (iii).

From (1) we see that the number of l for which (i) occurs is at most $x \exp(-(1/\sqrt{2}+o(1))\sqrt{\log x}\sqrt{\log\log x})$. From the union bound we see that the number of l for which (ii) occurs is at most

$$\ll \sum_{p > x^{1/u}} x/p^2 \ll x^{1-1/u} = x \exp(-(1/\sqrt{2}+o(1))\sqrt{\log x}\sqrt{\log\log x}).$$

Finally, from the pigeonhole principle we see that the number of l for which (iii) occurs is also at most

$$x^{1-1/u} = x \exp(-(1/\sqrt{2}+o(1))\sqrt{\log x}\sqrt{\log\log x}).$$

Thus one has $L(x) \ll x \exp(-(1/\sqrt{2}+o(1))\sqrt{\log x}\sqrt{\log\log x})$, as desired. This completes the proof.

The upper bound arguments seem more crude to the author than the lower bound arguments, so I conjecture that the lower bound is in fact the truth: $L(x) = x \exp(-(\sqrt{2} + o(1))\sqrt{\log x}\sqrt{\log\log x})$.

Comments (14)

Anonymous · 9 August 2024

first

allenknutson · 9 August 2024

I just felt like mentioning that that's the Zaharescu you and I went to grad school with.

Anonymous · 9 August 2024

There seems to be a citation missing in "... thanks to https://en.wikipedia.org/wiki/Siegel It turns out that this problem was essentially solved (though not explicitly) by a /href, who studied ...".
[Corrected, thanks – T.]

Anonymous · 10 August 2024

Write $L := \exp(100\sqrt{\log x})$.
Can we say that there are at most x/L choices of $y < x$ where $a_1, ..., a_t$ live in the interval $[y, y + L]$, $\prod_i a_i$ is a square, and $t < \exp(10^{-10}\sqrt{\log x})$? This sort of thing would get close to sharpening the bound.
The only non-obvious part is improving the bound for (iii) using this (hypothetical) information.

> **Anonymous** · 10 August 2024
>
> Hm. I guess to improve (iii), you could simply make u slightly larger. Even though that would hurt (i) slightly, the gain we'd hypothetically have from result about few y's would override this.
>
> > **Anonymous** · 10 August 2024
> >
> > Okay, if $a_1, ..., a_t$ live an interval of length L, then $\gcd(a_i, a_j) < L$ for distinct i, j it follows that that the "square-free" part s_i of each a_i should be bounded by L^{t-1} (since $s_i \mid \prod_{j \neq i} \gcd(a_i, a_j)$).
> > Thus any such interval must contain some element a of the form st^2 for $s < L^t < n^{1/100}$ (since $100 \cdot 10^{-10} < 1/100$). But the number of such a is bounded by $n^{1/100+1/2} < n/L^2$.
> > Each a belongs to at most L intervals, so we win. QED
> > Perhaps I will think about optimizing later. But someone else is welcome to do so.
> >
> > > **Anonymous** · 10 August 2024
> > >
> > > The non-constructive lower bound argument generalizes from squares to (say) cubes, or even weird requirements where you fix some $q_p > 1$ for each prime and ask for many products $n = \prod_i a_i$ where $q_p \mid \nu_p(n)$ for every prime p. You can define a multiplicative homomorphism f, now to a product of cyclic groups with sidelengths $q_2, q_3, ...$ You want to find some submultiset of $f(a_1), ..., f(a_t)$. I claim there exists some constant C depending only on $\mathrm{lcm}(q_{p_1}, ..., q_{p_d})$ (the torsion of the image of f), so that if $2^{t^{1/C}} > (\prod_{i \leq d} q_{p_i})$, then the multisubset exists.
> > > I have been told there is a slick proof of this via the polynomial method (which might even get $C = 1$ or something). But all I can remember is a long-winded elementary argument that gets $C = \log_2(\mathrm{lcm}(q_{p_1}, ...))$ (in particular for 2-torsion it recovers $C = 1$ as you obtain, but without having to even mention linear independence).

Anyways, I wonder what is the general behavior for more other choices of q_p. Probably this is just a mess, but perhaps it could be interesting.

Anonymous · 10 August 2024

Ah, my "improvement" forgot a factor of $sqrtloglogx$ in the exponent. Unfortunate. I guess it gives a decent bound without any smooth estimates. I will think a bit about fixing.



Terence Tao · 10 August 2024

I think this argument only works for t as large as $\sqrt{\log x}$ or so, rather than the $\exp(\sqrt{\log x})$ type levels that one would need to make a dent in the leading term in the asymptotics (as opposed to a lower order term).

Anonymous · 10 August 2024

ah yes, I did that very wrong. back to the drawing board.

Anonymous · 11 August 2024

Are the best constants in the exponents of the lower and upper bounds be necessarily the same? *[I see no reason to expect this – T.]*

Anonymous · 14 August 2024

How can they not be the same?



Terence Tao · 16 August 2024

Ah, I misinterpreted the question. Most likely the exponents are indeed the same (and as stated in the text, I believe they are both $\sqrt{2}$). However, it is theoretically possible that the bound oscillates infinitely often between two different exponents, as x goes to infinity.

Anonymous · 13 August 2024

The sentence after equation (2) is incomplete.